User and Item Prototypes (U-ProtoMF, I-ProtoMF)

SAMPLE v2 (thesis-write, 2026-09-09) | discussion material, not thesis prose.

Matrix factorization describes a user and an item by a vector of learned numbers whose coordinates stand for nothing in particular, so any account of why a recommendation was made has to be imposed on the model from the outside. ProtoMF [2] changes the representation instead of the account. It places a small set of learned reference vectors, the prototypes, into the latent space and describes every entity by how similar it is to each of them. A user is then no longer an unstructured vector but a profile over a handful of shared reference points, and since that profile is what the score is computed from, the reasoning of the model can be read off its own arithmetic rather than reconstructed afterwards.

This makes ProtoMF an instance of the interpretable-by-design models of Section 2.2, and it is the host the remainder of this work builds on, which is why we teach it at working depth. What the prototypes stand for is not fixed by the architecture, a point Section 2.5 returns to. The notation introduced in the following stays in use for the rest of the thesis.

Let $\mathcal{U} = \{u_i\}_{i=1}^{N}$ and $\mathcal{T} = \{t_j\}_{j=1}^{M}$ be the set of N users and M items of a system, and let $\mathcal{I} = \{(u_i, t_j)\}$ be the observed implicit interactions, where $(u_i, t_j) \in \mathcal{I}$ states that user u_i has interacted with item t_j. Indices are omitted whenever a statement holds for an arbitrary user or item. Matrix factorization assigns a learned embedding $\mathbf{u}$ to every user and a learned embedding $\mathbf{t}$ to every item and scores a pair by the dot product of the two. ProtoMF [2] keeps these embeddings and adds prototypes, which are further learned vectors placed in the same space as the entities they describe. The model is built in three steps, prototypes on the user side, prototypes on the item side, and both at once, and we follow that order.

U-ProtoMF rests on the assumption that the users of a system share a limited number of item consumption patterns, each of which can be embodied by a prototypical user. One user prototype may stand for the preference for Folk Metal tracks, another for the enjoyment of drama and romance movies [2]. The model introduces a set $\mathcal{P}^u$ of L^u user prototypes with $L^u \ll N$, each of them a learned vector $\mathbf{p}_l^u$ of the embedding dimension d. Every user then receives a second representation $\mathbf{u}^*$, the vector holding the similarity of the user embedding to each of the prototypes,

$$\mathbf{u}^* = \begin{pmatrix} \mathrm{sim}(\mathbf{u}, \mathbf{p}_1^u) \\ \vdots \\ \mathrm{sim}(\mathbf{u}, \mathbf{p}_{L^u}^u) \end{pmatrix} \in \mathbb{R}^{L^u}, \qquad \mathrm{sim}(\mathbf{a}, \mathbf{b}) = 1 + \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \cdot \|\mathbf{b}\|}, \tag{1}$$

where sim is the shifted cosine similarity and $\|\mathbf{x}\|$ is the L^2 norm of $\mathbf{x}$. The shift keeps every entry of $\mathbf{u}^*$ positive and inside the range from 0 to 2.

The affinity score of a user and an item is a linear combination of this new user representation and the item embedding,

$$\text{U-score}(u,t) = \sum_{l=1}^{L^u} s_l^{\text{user}}, \qquad \mathbf{s}^{\text{user}} = \mathbf{u}^* \odot \mathbf{t}, \tag{2}$$

where $\odot$ is the element-wise product and the item embedding $\mathbf{t} \in \mathbb{R}^{L^u}$ now carries one entry per user prototype. Equation 2 is the dot product $\text{U-score}(u,t) = \mathbf{u}^{*\top}\mathbf{t}$, written in a form that exposes the intermediate vector $\mathbf{s}^{\text{user}}$, which breaks the score down into one contribution per user prototype.

Three properties of this construction carry the model's interpretability claim. Every coordinate of $\mathbf{u}^*$ belongs to one prototype and therefore to one consumption characteristic, so a user is described one characteristic at a time. The item embedding acts as the vector of weights an item places on those characteristics, and a heavy metal track is expected to carry a high weight on the coordinate of the prototype that represents metal fans. The score is a linear function of the similarities, so every prototype holds a quantifiable share of a given recommendation. Two further points follow from the shape of $\mathbf{u}^*$. The representation is graded rather than exclusive, since a user is related to all prototypes at once instead of being assigned to one of them, which allows several tastes to coexist in one profile. And that profile is not a description placed beside the model, it is the representation the score is computed from. Anchoring a latent space with a small set of shared vectors is not unique to ProtoMF, and anchor-based collaborative filtering [1] is its closest relative, using one shared set of anchors for both users and items.

I-ProtoMF applies the same construction to the item side, where the assumption is that items are co-consumed in a limited number of patterns, so that an item prototype represents, for example, the items of one musical genre or product category. The model learns a set $\mathcal{P}^t$ of L^t item prototypes $\mathbf{p}_l^t$ with $L^t \ll M$ and represents every item by its similarities to them,

$$\mathbf{t}^* = \begin{pmatrix} \text{sim}(\mathbf{t}, \mathbf{p}_1^t) \\ \vdots \\ \text{sim}(\mathbf{t}, \mathbf{p}_{L^t}^t) \end{pmatrix} \in \mathbb{R}^{L^t}. \tag{3}$$

The score mirrors Equation 2,

$$\text{I-score}(u,t) = \sum_{l=1}^{L^t} s_l^{\text{item}}, \qquad \mathbf{s}^{\text{item}} = \mathbf{t}^* \odot \mathbf{u}, \tag{4}$$

where the user embedding $\mathbf{u} \in \mathbb{R}^{L^t}$ now carries one entry per item prototype and the score is again a dot product, $\text{I-score}(u,t) = \mathbf{u}^\top \mathbf{t}^*$. The element-wise

form serves the same purpose as before and splits the score over the item prototypes.

Both models are trained on the observed interactions with the cross-entropy loss over the softmax of the scores,

$$\mathcal{L}_{rec}(\mathcal{I}, \Theta) = -\sum_{(u,t)\in\mathcal{I}} \ln p(t \mid u) + \lambda_{L2}\|\Theta\|, \qquad p(t \mid u) = \frac{e^{\text{U-score}(u,t)}}{\sum_{j=1}^{M} e^{\text{U-score}(u,t_j)}}, \tag{5}$$

where Θ holds the model parameters and λ_{L2} weights the L^2 regularization term. The loss of I-ProtoMF is the same with I-score in place of U-score. ProtoMF adds two further terms to this objective, which act on the geometry of the prototype space and are treated in Section 2.3.3.

The prototypes themselves are learned parameters and carry no name of their own, so both models are read after training, and the two sides are read differently. Item prototypes share their space with the item embeddings, so an item prototype is interpreted through its nearest items, and whatever their metadata has in common becomes its name. A user prototype cannot be read this way, because the objects a person recognizes are items rather than users. It is interpreted instead by building a synthetic representation $\mathbf{u}^*$ that assigns the maximum similarity to the prototype in question and zero to all others, and by reading the recommendations the model returns for that imaginary user [2]. In both cases the name of a prototype is a summary of the neighborhood it sits in, obtained from the trained model by inspection.

References

[1] O. Barkan, R. Hirsch, O. Katz, A. Caciularu, and N. Koenigstein. Anchor-based collaborative filtering. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 2877–2881, 2021.

[2] A. B. Melchiorre, N. Rekabsaz, C. Ganhör, and M. Schedl. Protomf: Prototype-based matrix factorization for effective and explainable recommendations. In *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys)*, pages 246–256, 2022.